

Ethics, governance, and equity

Data Science for Health Systems | Washa Takwimu 2026 Short Course | Day 1

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40 minutes: concepts, two frameworks, one case

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|---|-------|---|--------|
| ● | 13:30 | Welcome and the case we will follow | 5 min |
| ● | 13:35 | Why ethics and governance matter for routine data | 3 min |
| ● | 13:38 | Six concepts, applied to the case | 10 min |
| ● | 13:48 | Worked examples: Tanzania and South Africa | 7 min |
| ● | 13:55 | Case discussion in groups | 7 min |
| ● | 14:02 | Questions and open discussion | 7 min |
| ● | 14:09 | Bridge to R Practical 1 | 1 min |

Two discussion blocks

14 minutes in total for the group case and open questions.

Not legal advice

Laws are used as worked examples. Check current rules and your institution for real projects.

By the end of this session you will be able to:

-  Explain six ideas that shape the ethical use of health-system data: lawful basis, purpose limitation, ownership and stewardship, community interests, reporting incentives and unintended harms.
-  Locate the main rules for personal health data in Tanzania and South Africa, and see what they have in common.
-  Apply these ideas to a realistic data-sharing request and say what you would ask before saying yes.

DHIS2 Overview

DHIS2 is an open source, web-based platform for data collection, management, visualization, and integration. It is most commonly used in the health sector as a health management information system (HMIS)—and increasingly for electronic health records—but it is usable in any sector due to its flexible, customizable design and its powerful set of modular software features. It enables users and policy makers to generate analysis from live data in real time, and is designed to function even in environments where internet access is limited.

Today, DHIS2 is the world's largest HMIS platform, used by 80+ low and middle-income countries (LMICs) as the government-wide system of record for health data. Including NGO-based programs, DHIS2 is in use in more than 100 countries. This has made a significant impact, helping transform public sector governance and use of data to achieve outcomes in health, education and sustainable development goals.

The DHIS2 core platform is developed and maintained through a global collaboration managed by the HISP Centre at the University of Oslo (HISP UiO), supported by a coalition of partners and investors. HISP is a global network comprised of 24 local HISP groups in LMICs that help ensure that DHIS2 meets the needs of local users and provide long-term, hands-on DHIS2 technical assistance, capacity building, and support to government partners and other organizations.



About DHIS2

DHIS2 is a digital public good developed by HISP UiO, in collaboration with a global network of partners, that powers health information management around the world and is transforming information systems across sectors—from education and logistics to climate and beyond.

A request for HIV testing data

A fictional scenario

A regional health management team holds monthly facility reports in DHIS2 and patient-level HIV testing registers from 60 facilities.

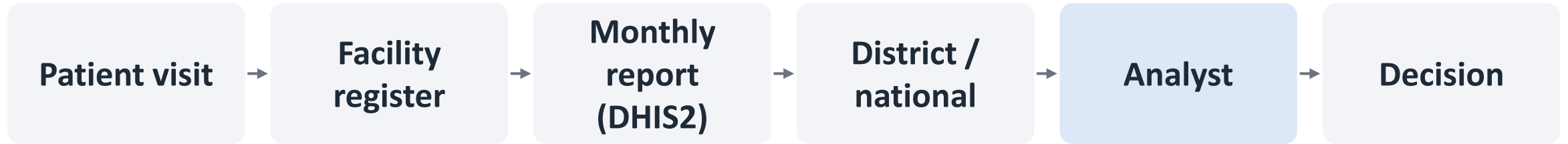
A university partnered with a private technology company asks for patient-level records (age, sex, village, test result, etc.) to build a model that predicts where to send outreach teams. The company's servers are outside the country.

Facilities receive results-based payments for the number of people tested. The team also plans to publish a ranking of facilities.

Keep asking:

- On what basis can these data be used?
- Is this the purpose they were collected for?
- Who decides: patients, facilities, the ministry?
- Could communities be helped or harmed?
- What do the payments do to the numbers?
- What could go wrong once the data leave?

Every step of the data journey carries duties



“Anonymised” is not the same as anonymous

99.98%

of Americans could be correctly re-identified in an “anonymised” dataset using 15 demographic attributes such as age, sex, and marital status (Rocher et al., Nature Communications, 2019).

Health-system data are collected for care and management, by people who trust the system. Analysis is a later, separate use that needs its own justification.

Sources: [Rocher, Hendrickx & de Montjoye 2019](#)

Six questions to ask of any health data use



Lawful basis

On what legal ground is this processing allowed?



Purpose limitation

Is the new use compatible with why the data were collected?



Ownership and stewardship

Who has rights over the data, and who is accountable?



Community interests

Could a group benefit or be harmed, even if no individual is named?



Reporting incentives

Do targets or payments change what gets recorded?



Unintended harms

What could go wrong after release: re-identification, stigma, misuse?

Health data are ‘sensitive’: the bar is higher

	Tanzania: Personal Data Protection Act (PDPA 2022)	South Africa: Protection of Personal Information Act (POPIA 2013)
Health is...	“Sensitive personal data” (s3)	“Special personal information” (s26)
Default rule	No processing without prior written consent (s30)	Processing prohibited unless an s27 or later exception applies
Exceptions relevant to us	Scientific research where the Commission has set guidelines; medical purposes under a health professional (s30)	Research in the public interest, or where consent is impossible or disproportionate with sufficient privacy guarantees (s27(1)(d)); care and administration by health professionals and institutions (s32)

In the case: patients consented to testing, not to a company’s model. A research exception may apply, but only with ethics approval and safeguards, not by default.

Sources: [Tanzania PDPA 2022](#) | [South Africa POPIA](#)

Collected for one purpose, used for another?

Both laws start from the same principle: collect for a specific, explicit purpose (Tanzania PDPA s5; POPIA s13).

Tanzania: s25(2)

Further use is allowed, among other grounds, if the data are used in a form where the person is not identified, or for statistical or research purposes and not published in an identifiable form.

South Africa: s15

Further processing must be compatible with the original purpose. It is compatible if used solely for historical, statistical or research purposes and not published in an identifiable form (s15(3)(e)).

In the case: a public ranking and a commercial product are not obviously ‘solely research’, and in Tanzania commercial use needs the person’s consent. Aggregated, non-identifiable data would be much easier to justify.

Sources: [Tanzania PDPA 2022 s5, s25](#) | [POPIA s13, s15](#)

Who ‘owns’ health data? Who is accountable?

- Data protection laws do not give anyone simple ‘ownership’. They give people rights over their data and place duties on those who hold it: ‘data controllers’ (Tanzania) or ‘responsible parties’ (South Africa).
- Facilities, districts and ministries are stewards: they decide on access and answer for misuse.
- Moving data across borders needs extra steps: a permit from the Personal Data Protection Commission in Tanzania (s31–32); adequate protection or another s72 ground in South Africa.

Continental context

The AU Data Policy Framework (July 2022) lists sovereignty, fairness and inclusiveness among its principles: countries should be able to govern and benefit from their own data.

The AU Malabo Convention on personal data protection entered into force on 8 June 2023.

In the case: foreign servers trigger cross-border rules. Who keeps the model and the derived data?

Sources: [PDPA s31–32](#) | [POPIA \(s72\)](#) | [AU Data Policy Framework \(FPF\)](#) | [Malabo Convention \(CIPIT\)](#)

Harm can fall on a group even when no one is named

- Aggregated results about a village, ethnic group or key population can stigmatise everyone in it.
- Communities often carry the burden of data collection but see few of the benefits.
- CIOMS Guideline 7: Communities must be engaged in the planning, conduct, and dissemination of research.

CARE principles (GIDA, 2020)

- C** Collective benefit
- A** Authority to control
- R** Responsibility
- E** Ethics

Written for Indigenous data; widely used as a lens for collective interests.

In the case: village-level predictions of ‘high HIV risk’ could stigmatise villages. Were communities consulted? Who benefits?

Sources: [CIOMS 2016 guidelines](#) | [CARE principles \(Imperial summary; Carroll et al. 2020\)](#)

When a number becomes a target, it changes

Evidence from results-based financing in Zimbabwe (voluntary medical male circumcision, community verification):

19%

of interviewed patients' records were classified as over-reported (263 of 1,412)

2.9×

higher odds of over-reporting for clients aged 15–29, the age group that earned higher payments (OR 2.92)

45%

of sampled patients could not be interviewed at all (1,153 of 2,565)

Payments, targets and rankings can shape what is recorded. This is a data-quality problem and an ethical one: decisions and resources follow the numbers.

In the case: facilities are paid per person tested. A model trained on these counts may learn reporting behaviour, not need.

Sources: [Gorman et al., BMC Health Services Research 2025](#)

What could go wrong after the analysis?

Re-identification

Small numbers plus a few attributes can identify a person. Suppress or combine small cells before sharing.

Stigma and labelling

Rankings and 'hotspot' maps can label facilities or areas as failing, or communities as diseased.

Misuse by others

Data on criminalised or marginalised groups can put people at risk if they reach the wrong hands.

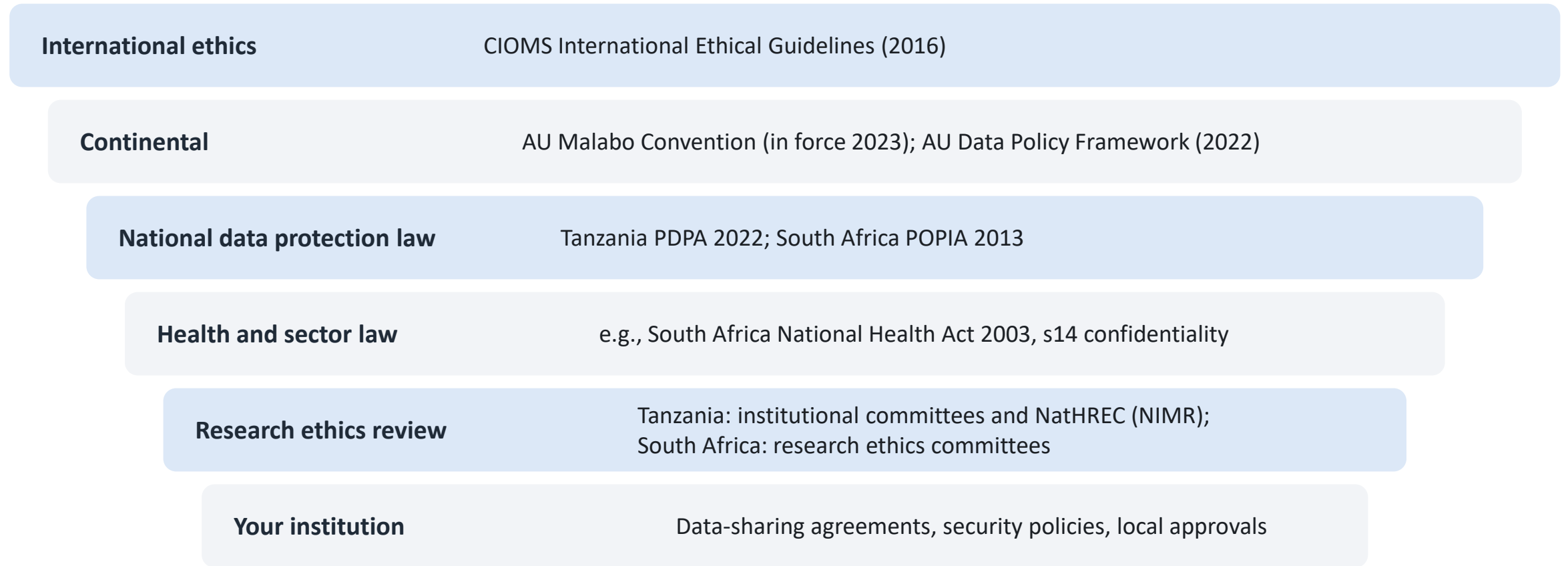
Lost trust

If people fear disclosure, they may avoid testing and care, which harms the health system itself.

In the case: a facility with 12 positive tests in a month and a published ranking: both risks at once.

Sources: [Rocher et al. 2019](#) | [UNAIDS guidelines on HIV information confidentiality and security \(2007\)](#)

Rules come in layers



Sources: [CIOMS 2016](#) | [Malabo \(CIPIT\)](#) | [AU DPF \(FPE\)](#) | [NHA s14](#) | [NIMR](#)

Personal Data Protection Act, 2022

In force	1 May 2023 (Government Notice 326 of 2023)
Regulator	Personal Data Protection Commission (PDPC), fully operational since 3 April 2024
Duties	Controllers and processors register (valid 5 years) and appoint a Data Protection Officer
Principles (s5)	Lawful, fair, transparent; specific purpose; adequate and necessary; accurate; kept no longer than needed; secure
Health data	Sensitive personal data (s3); prior written consent unless an exception applies (s30)
Further use (s25)	Allowed if not identifiable, or for statistics and research without identifiable publication
Cross-border (s31–32)	PDPC permit, plus adequate protection or safeguards in the recipient country
Automated decisions	Right not to be subject to solely automated decisions; controllers must explain the logic
Research ethics	Institutional ethics approval, or NatHREC where there is none; COSTECH permit for non-Tanzanian researchers

Sources: [PDPA 2022](#) | [Commencement \(Afriwise\)](#) | [NIMR research regulations](#)

POPIA 2013 and the National Health Act 2003

In force	Main provisions from 1 July 2020, with one year to comply (to 30 June 2021)
Regulator	Information Regulator
8 conditions (s4)	Accountability; processing limitation; purpose specification; further processing limitation; information quality; openness; security safeguards; data subject participation
Health data	Special personal information (s26); research exception s27(1)(d); health professionals and institutions s32
Confidentiality	National Health Act s14: user information is confidential; disclosure only with written consent, a court order or law, or a serious public-health threat
Recent	Health-information regulations for insurers, medical schemes and employers, in force 6 March 2026. ASSAf's voluntary POPIA Compliance Framework for Research, endorsed May 2025

Sources: [POPIA](#) | [NHA s14](#) | [2026 health regulations \(Moonstone\)](#) | [ASSAf framework](#)

Different laws, very similar questions

Question for the case	Tanzania	South Africa
Is HIV status a special category?	Yes: sensitive personal data	Yes: special personal information
Research use without consent?	Only via the scientific-research exception and PDPC guidelines	Only via s27(1)(d) with safeguards
Re-use if not identifiable?	Yes, s25(2)	Yes, s15(3)(e)
Servers abroad?	Commission permission and safeguards	s72 conditions
Ethics review?	Institutional committee or NatHREC	Research ethics committee
Decisions made by the model?	No solely automated decisions with significant effects; explain the logic	s71 limits decisions based solely on automated profiling

Common thread: the less identifiable and the more clearly public-interest the use, the easier it is to justify.

Sources: [PDPA 2022](#) | [POPIA](#)

In your group: would you share these data?



Groups of 4–5. Choose one question each. Be ready to share your thoughts.

- A** What would you need before sharing anything: consent, ethics approval, a data-sharing agreement, a permit?
- B** Could the purpose be met with aggregated facility-month data instead of patient records?
- C** Who should be consulted: facility staff, communities, people living with HIV?
- D** How might results-based payments bias the model, and what would you check?
- E** Should the facility ranking be published? What would you show instead?

You will see these ideas in today's practical

Synthetic data

No real patients, so we can share it with the whole class.

Identifiers removed

Names, ID numbers and addresses were dropped before you received the file: data minimisation.

Purpose written down

Each variable has a defined meaning and source in the data dictionary.

Small numbers flagged

Counties with fewer than 100 patients are marked: precision and re-identification risk.



Please open RStudio and your r-practical-1 Rmd now.